MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, ADJUSTABLE, PRECISION VOLTAGE REFERENCE,
SHUNT REGULATOR, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/148A, dated 5 September 1989, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

Paragraph 3.1, delete and substitute the following:

" 3.1 Item requirements. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein. The individual item requirements for devices marked with the "Q" certification mark shall be in accordance with MIL-I-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein. "

PAGE 3

Table I, at the end of the title, put footnote " 1/ ". At the end of the table, write "See footnote at end of table."

PAGE 4

Table I, at the end of the title, put footnote " 1/ ". At the end of the table, write " 1/ For devices marked with the "Q" certification mark, the parameters listed herein may be guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan. "

TABLE I, Reference adjustment ($V_Z$ADJ), device type 02 only, delete and substitute the following:

"

| | | | | | |
|---|---|---|---|---|---|
| Reference adjustment | $V_{ZADJ}$ | | 100.0 | | mV |

"

TABLE I, Delta voltage/delta temperature ( $V_Z$/dT ), device type 02 only, delete and substitute the following:

"

| | | | | | |
|---|---|---|---|---|---|
| Delta voltage/ delta temperature | $\frac{V_Z}{dT}$ | From -55°C to +25°C<br>From +25°C to +125°C<br>$I_R$ = 1 mA | 15<br>15 | | mV |

"


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FIGURE 6, delete and substitute the following:

"

Device type 02

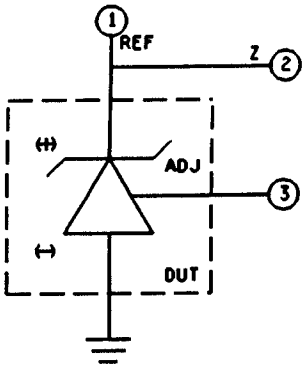



FIGURE 6. Test circuit - Continued.

"



FIGURE 7, device type 02, note 2, delete and substitute the following:

" 2. $V_{CC}$ = +20 V +0 V
+2 V "

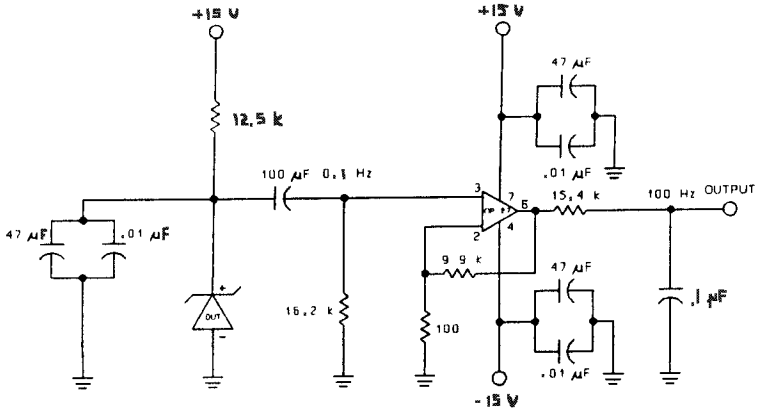


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FIGURE 8, delete and substitute the following:

"




NOTES:
1. Test time = 10 seconds.
2. $V_{OUT}$ measured with differential amplifier 7A22 and lower frequency set to 0.1 Hz.
3. Device noise = $V_{OUT}$ / 100

FIGURE 8. Noise test circuit for device type 02 - Continued.

"

PAGE 14

Table III, at the end of the title, put footnote " 1/ ".

Table III, subgroup 1, $T_A$ = 25°C, $V_{KA10}$, notes column, renumber footnote " 1/ " to footnote " 2/ ".

Table III, subgroup 1, $T_A$ = 25°C, $V_{KA36}$, notes column, renumber footnote " 1/ " to footnote " 2/ ".

Table III, subgroup 1, $T_A$ = 25°C, $Z_{KA}$, notes column, renumber footnote " 2/ " to footnote " 3/ ".

Table III, subgroup 2, $T_A$ = 125°C, $V_{KA10}$, notes column, renumber footnote " 1/ " to footnote " 2/ ".

Table III, subgroup 2, $T_A$ = 125°C, $V_{KA36}$, notes column, renumber footnote " 1/ " to footnote " 2/ ".

Table III, subgroup 2, $T_A$ = 125°C, $Z_{KA}$, notes column, renumber footnote " 2/ " to footnote " 3/ ".

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Table III, at the end of the title, put footnote " 1/ ".

Table III, subgroup 3, $T_A$ = -55°C, $V_{KA10}$, notes column, renumber footnote " 1/ " to footnote " 2/ ".

Table III, subgroup 3, $T_A$ = -55°C, $V_{KA36}$, notes column, renumber footnote " 1/ " to footnote " 2/ ".

Table III, subgroup 3, $T_A$ = -55°C, $Z_{KA}$, notes column, renumber footnote " 2/ " to footnote " 3/ ".

TABLE III, Test no. 28, notes column: add "4/".



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Bottom of page, footnote "1/" should state "1/ For devices marked with the "Q" certification mark, the parameters listed herein may be guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

Bottom of page, renumber footnote "1/" to "2/".

Bottom of page, renumber footnote "2/" to "3/".

Bottom of page, below footnote "3/", add footnote "4/" as follows:

"4/ If not tested, shall be guaranteed to the limits specified in table III herein."

PAGE 16

TABLE III, Subgroup 1, $T_A$ = +25°C, delete and substitute the following:

"

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $V_{ADJ}$ | 6 | V4 | 1 mA | | 600 mV | 2 | $V_{ADJ}$ = V4 | | 2.500 | V |
| | 7 | V5 | 1 mA | | 1.9 V | 2 | $V_{ADJ}$ = V5 | 2.500 | | |
| $V_{ZADJ}$ | 8 | | | | | 2 | $V_{ZADJ}$ = V5 - V4 | 100.0 | | mV |

"

TABLE III, Subgroup 1, $T_A$ = +25°C, $V_Z/I_Z$, Test no. 9, reference column, delete and substitute the following:

" 0.4 mA "

TABLE III, Subgroup 1, $T_A$ = +25°C, $V_Z/I_Z$, equations column, delete and substitute the following:

" $V_Z/I_Z$ = |(V6 - V7)/9.6 mA| "

TABLE III, Subgroup 2, $T_A$ = +125°C, delete and substitute the following:

"

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $V_{ADJ}$ | 16 | V11 | 1 mA | | 600 mV | 2 | $V_{ADJ}$ = V11 | | 2.500 | V |
| | 17 | V12 | 1 mA | | 1.9 V | 2 | $V_{ADJ}$ = V12 | 2.500 | | |
| $V_{ZADJ}$ | 18 | | | | | 2 | $V_{ZADJ}$ = V12 - V11 | 100.0 | | mV |

"

TABLE III, Subgroup 2, $T_A$ = +125°C, $V_Z/I_Z$, Test no. 19, reference column, delete and substitute the following:

" 400 µA "

TABLE III, Subgroup 2, $T_A$ = +125°C, $V_Z/I_Z$, equations column, delete and substitute the following:

" $V_Z/I_Z$ = |(V14 - V13)/9.6 mA| "



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TABLE III, Subgroup 3, $T_A$ = -55°C, delete and substitute the following:

"

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $V_{ADJ}$ | 26 | V18 | 1 mA | | 600 mV | 2 | $V_{ADJ}$ = V18 | | 2.500 | V |
| | 27 | V19 | 1 mA | | 1.9 V | 2 | $V_{ADJ}$ = V19 | 2.500 | | |
| $V_{ZADJ}$ | 28 | | | | | 2 | $V_{ZADJ}$ = V19 - V18 | 100.0 | | mV |

"

TABLE III, Subgroup 3, $T_A$ = -55°C, $V_Z/I_Z$, Test no. 29, reference column, delete and substitute the following:

" 400 μA "

TABLE III, Subgroup 3, $T_A$ = -55°C, $V_Z/I_Z$, equations column, delete and substitute the following:

" $V_Z/I_Z$ = |(V21 - V20)/9.6 mA| "

TABLE III, Subgroup 4, $T_A$ = +25°C, $N_O$, equations column, delete and substitute the following:

" See figure 8 "

TABLE III, Subgroups 5 and 6, delete and substitute the following:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $V_Z/dt$ | 32 | | | $V_Z/dt$ = \|V9 - V2\| | | 15 | mV |
| $V_Z/dt$ | 33 | | | $V_Z/dt$ = \|V16 - V2\| | | 15 | mV |

Bottom of page, to footnote "1/" add the following: "For devices marked with the "Q" certification mark, the parameters listed herein may be guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

PAGE 18

Paragraph 4.1, delete and substitute the following:

" 4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510, and methods 5004, 5005, and 5007, as applicable, of MIL-STD-883, except as modified herein. For devices marked with the "Q" certification mark, sampling and inspection procedures shall be in accordance with MIL-I-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein. "

Paragraph 4.2a(1), delete and substitute the following:

" (1) For class S devices, test condition C, using the circuit shown of figure 7. "

Paragraph 4.2a(2), delete and substitute the following:

" (1) For class B devices, test condition C, using the circuit shown of figure 7. "



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Paragraph 4.2, delete the following:

" NOTE: If accelerated high-temperature test conditions are used, the device manufacturer shall ensure that at least 85 percent of the applied voltage is dropped across the device at temperature. The device is not considered functional under accelerated test conditions. "

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Paragraph 4.4.3b(1), delete and substitute the following:

" (1) Test condition C, using the circuit shown on figure 7. "

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH, TD
Air Force - 19, 85, 99
DLA - ES

Civil Agency Coordinating Activity:
NASA - NA

Preparing activity:
DLA - ES

Agent:
DLA - ES

(Project 5962-1393)



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